Activity 2: Run a Controlled Vibe-Coding Change

Course: Build ASP.NET Web Apps with Vibe Coding (TGS-2023040477)

Alignment: LO1 · K1/A1

Duration: 45 minutes

Runnable snapshot: src/LocalCRM

Goal

Add a health endpoint through a specify–generate–inspect–verify loop without accepting unrelated AI edits.

What you will produce

a reviewed /health endpoint, build evidence and a decision log

Safety and data boundary

Use only the supplied fictional records in mock-data/. Do not paste personal, customer, credential, or production data into an AI tool. Review every generated diff before running it.

Vibe script

> Modify LocalCRM/Program.cs only. Add GET /health returning HTTP 200 JSON { status: 'ok', service: 'LocalCRM' }. Do not add packages or reorder existing middleware. Explain the exact diff and provide curl acceptance evidence.

Step-by-step

Step 1 — Confirm the baseline

Work from the Activity 1 commit and make sure the tree is clean.

```
git status --short
dotnet build LocalCRM/LocalCRM.csproj
```

Step 2 — Submit the constrained prompt

Use the provided prompt with your approved coding assistant. Reject any response that adds packages or edits more than Program.cs.

Step 3 — Apply the smallest patch

Place the endpoint after app.UseAuthorization() and before app.MapRazorPages().

Create or update Program.cs excerpt:

```
app.UseAuthorization();

app.MapGet("/health", () => Results.Ok(new
{
    status = "ok",
    service = "LocalCRM"
}));

app.MapRazorPages();
```

Step 4 — Inspect scope

Confirm exactly one source file changed and no package reference was added.

```
git diff --stat
git diff -- LocalCRM/Program.cs
git diff -- LocalCRM/LocalCRM.csproj
```

Step 5 — Build and run

Compile and start the application on the known URL.

```
dotnet build LocalCRM/LocalCRM.csproj
dotnet run --project LocalCRM --urls http://localhost:5050
```

Step 6 — Capture positive evidence

Probe headers and JSON from another terminal.

```
curl -i http://localhost:5050/health
```

Expected evidence: HTTP 200, application/json, status=ok and service=LocalCRM.

Step 7 — Capture negative evidence

Probe a misspelled path to prove routing distinguishes unmatched endpoints.

```
curl -i http://localhost:5050/healthz
```

Expected evidence: HTTP 404.

Step 8 — Record the decision

Create evidence/activity-02.md with prompt, reviewed files, commands and observed status codes; then commit.

```
mkdir -p evidence
git add LocalCRM/Program.cs evidence/activity-02.md
git commit -m 'feat: add verified health endpoint'
```

Run the supplied snapshot

The Activity folder already contains a complete stage-appropriate ASP.NET Core project and mock data. From this Activity folder:

```
cd src/LocalCRM
dotnet restore
dotnet run --urls http://localhost:5102
```

Open http://localhost:5102. For Activities 7–10, also probe http://localhost:5102/api/contacts; for Activities 8–10, probe /openapi/v1.json and /health/ready.

Acceptance checklist

□ The project targets net10.0 and restores without an unreviewed package.

□ The app starts from this Activity folder without relying on another Activity.

□ The supplied mock contacts are fictional and load as described.

□ Positive and negative paths match the expected status or UI state.

□ The diff contains only intended files and no secret.

Troubleshooting

- NETSDK1045: install a .NET 10 SDK or select the installed 10.0 feature band in global.json.
- Address already in use: stop the previous Activity process or use --urls http://localhost:PORT with a free port.

- HTTP 404: compare the exact route, method, route constraint and Activity stage.
- HTTP 400/415: inspect ProblemDetails and confirm JSON shape plus Content-Type: application/json.
- SQLite failure (Activities 8–10): remove only the local training localcrm.db, then restart so the snapshot recreates mock data.

Reflection and evidence record

1. Explain how the evidence demonstrates LO1 and K1/A1.

2. Name one AI suggestion you rejected or changed and why.

3. Identify one negative path you tested and the observable result.

4. State the next improvement you would make without widening the approved scope.

Evidence item	Record the path, command result or observation
Build or test result	
Runtime URL and HTTP status	
Positive-path evidence	
Negative-path evidence	
Reviewed AI change or rejection	
Commit or evidence reference	